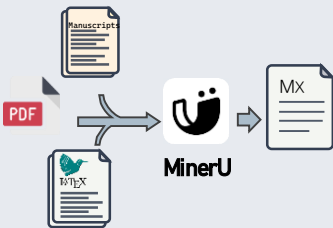


File-Conversion Module



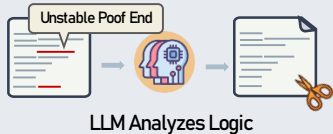
- Convert diverse sources into Markdown
- Maintain mathematical expressions

Regex & Prompt Refinement

Captured Elements:

Theorems, Assumptions, Lemmas, Conditions, Propositions Definitions

LLM Semantic Boundary Resolution



- Corrects Delimiter Drops, Clear Redundant Transitions

Context-Segmentation Engine

(1) Core Element Extraction

Theorem 1, Assumption 1, Definitions, Extract math Statements & premises Via Regex

(2) Implicit Proofs Handling

Large proof, Proof, Nested proofs handled, Leaves markers

(3) Equation Extraction

Tags, Tags equations, Equations tags, Capture multi-line equations & tags

(4) Global Notation/Context

Global Symbol Dictionary & Background Capture, LLM processes

(5) Semantic Boundary Resolution

Correct proof ends, clear redundant sentences, Zero-deletion policy, LLM processes

Dependency Parsing Module

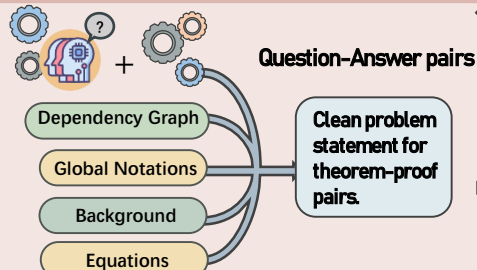
Internal Dependency Extraction

Lemma 2, Lemma 1, Theorem 2, Assumption A, Theorem 1, Topological dependency graph

External Citation Mapping

Map specific or vague citations for cross-document tracking

Base Problem Generation Module



- Only extract globally defined symbols, excluding symbols generated during proof.
- Construct semantically fluent and logically coherent setup.

Multi-Difficulty QA Generation

Non-Refine Mode

Direct mechanical assembly preserving absolute original logic and data fidelity

Refine Mode

AI-driven semantic optimization for coherent and properly structured reasoning.

Easy: Provided Facts Only, Medium: Prove Prerequisites First, Hard: Independent Chain Construction

Multi-Stage Quality Validation

